



Composing Context Compression for Long-Horizon LLM Agents

Controlled Composition and Critical-Fact Retention on top of ACON

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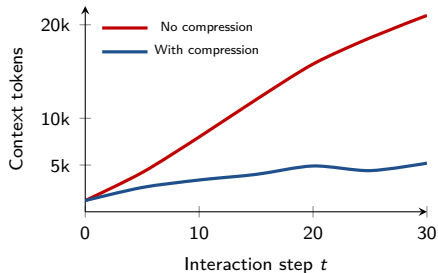
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Outline

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The Problem: Context Grows Without Bound



Two bottlenecks

- **Memory** — the KV cache scales with sequence length
- **Distraction** — long, partly stale context degrades decisions

The tension

Compression must be *aggressive* (to save tokens) and *precise* (one lost file path or API token derails an entire trajectory).

The Two Methods We Build On

ACON — prompt-level, generative

- An LLM *rewrites* the history when $|h_t| > T_{\text{hist}}$ and shortens observations when $|o_t| > T_{\text{obs}}$
- The compression guideline is optimised *in natural language*: contrast runs that succeed uncompressed but fail compressed
- Agent weights stay frozen; compressor is distilled into a small model
- AppWorld: 56.5% vs. 56.0% accuracy, peak tokens ↓ 26–54%

SAC — KV-level, soft

- Selects *anchor tokens*, marks them with a learned embedding
- Bidirectional attention lets anchor KV pairs absorb the whole context
- No autoencoding pretraining required
- MRQA, Llama-3.2-1B at 15×: 54.95 in-domain F1 (vs. EPL 51.52)

Our project **reproduces both on our own hardware**, then asks how compression stages should be *composed* and *evaluated*.

Where the Field Stands

Property	ACON	SAC	SWE-Pruner	LLMLingua	MEM1/MemAgent	Proposed
Output form	Text	KV cache	Text lines	Text tokens	Text memory	Text
Agent weights frozen	Yes	Yes	Yes	Yes	No	Yes
History compression	Yes	No	No	Yes	Merged	Yes
Observation compression	Yes	No	Yes	Yes	Merged	Yes
Composition studied	Partly	No	No	No	No	Yes
Intrinsic quality metric	No	No	No	No	No	Yes (CFR)

- “Partly”: ACON reports a combined setting in an appendix ablation only.
- “Merged”: a single learned memory holds both history and observations.
- Every existing method compresses *one* kind of context — how the stages **interact** is unstudied.

Experimental Setup

Hardware & serving

- One NVIDIA RTX 2000 Ada (16 GB)
- Models served locally through Ollama
- Agent: qwen2.5:14b
- Compressor: qwen2.5:7b

Benchmarks

- **ACON:** AppWorld train_history_tiny (38 tasks), seed 42, temperature 0
- **SAC:** authors' Llama-3.2-1B checkpoints on MRQA

Deviations from the papers — and why

- Thresholds $T_{\text{hist}}=512$, $T_{\text{obs}}=256$ instead of 4096/1024 — the paper values almost never fire on this split
- Ollama served both models with a **4096-token** context, so long baseline prompts were truncated
- One seed per setting so far

These caveats are carried explicitly into every claim that follows.

Result 1: ACON Reproduction on AppWorld

Run	Tasks	Max iter.	Solved	Avg iter.	Agent input tok.	Wall clock (s)
Baseline, no compression	8	12	1	11.00	269,935	953.9
History compression ($T_{\text{hist}}=512$)	8	12	2	10.75	232,749	882.8
Observation compression ($T_{\text{obs}}=256$)	8	12	1	11.00	261,422	873.6
Baseline, full split	38	30	12 (31.6%)	23.7	2,972,376	7312
Baseline, matched subset	18	30	7	22.22	1,283,906	—
History compression, matched subset	18	30	9	19.50	969,389	—

- On the 18 matched tasks: **9 solved vs. 7**, with **24.5% fewer agent input tokens**.
- The change is *not uniform* — it solved 5 tasks the baseline failed, and failed 3 the baseline solved.
- At this scale the latency overhead ACON reports did **not** appear; all compressed runs finished sooner.

Result 2: SAC Reproduction and the SAC $\times$ ACON Pilot

SAC reproduces to within 0.2 F1

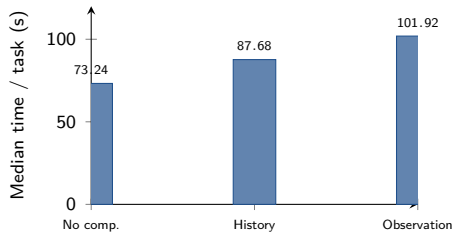
15 $\times$: **54.98** ID / **39.25** OOD (published 54.95 / 39.26)

5 $\times$: **63.70** / **47.71** (published 63.63 / 47.72)

Run	ID F1	OOD F1	Δ ID vs. R0 (95% CI)	Δ OOD vs. R0 (95% CI)
R0: SAC fine-tuning (matched control)	51.37	38.62	control	control
R1: + uniform KL (ablation)	51.06	38.17	-0.32 [-1.02, +0.44]	-0.45 [-2.11, +1.44]
R2: + FGD (gated KL, $\lambda=1$)	51.15	38.47	-0.23 [-0.95, +0.47]	-0.14 [-2.04, +1.77]
R3: + TGA + FGD	60.03	44.71	+8.66 [+7.66, +9.60]	+6.10 [+3.62, +8.96]
<i>Published SAC 15$\times$ (20,000 steps)</i>	<i>54.95</i>	<i>39.26</i>		

- **FGD is a null result.** Its intervals contain zero. A diagnostic explains why: the zero-shot full-context teacher had *higher* gold-answer NLL than the compressed student on **90.5%** of 2,000 training examples — there is little to distil.
- **TGA + FGD wins**, beating the published 20,000-step checkpoint after a quarter of the budget — but the question is seen during compression, so the memory is *not reusable* across questions. This is a separate query-aware setting, not a drop-in improvement.

Gap (a): Token Savings Do Not Become Time Savings



ACON's own measurement: Qwen3-14B agent & compressor, one A100.

Why compression can cost time

- The compressor generates **autoregressively** — it is itself an LLM call
- Rewriting the history **invalidates the agent's KV cache**, so the compressed prefix must be re-encoded from scratch

Consequence

A token-count objective $C(\pi)$ does not capture this overhead at all. Latency must be modelled as a *separate* term.

Gap (b): The Combined Setting Is Schedule-Confounded

The trigger set depends on the pipeline itself

$$\mathcal{K}(\pi) = \{ t : |h'_{t-1} \oplus (o'_t, a_t)| > T_{\text{hist}} \}$$

Since $|o'_t| \leq |o_t|$, compressing observations first makes $|h_t|$ grow more slowly. History compression then fires at **different steps**, and on **already-pruned text**: in general $\mathcal{K}(\pi_{ho}) \neq \mathcal{K}(\pi_h)$.

The interaction term

$$I = \Delta(\pi_{ho}) - \Delta(\pi_h) - \Delta(\pi_o)$$

On AppWorld: $\Delta(\pi_h) = +0.5$, $\Delta(\pi_o) = -2.4$ and $\Delta(\pi_{ho}) = -11.4$ points, so

$$I = -9.5 \text{ points.}$$

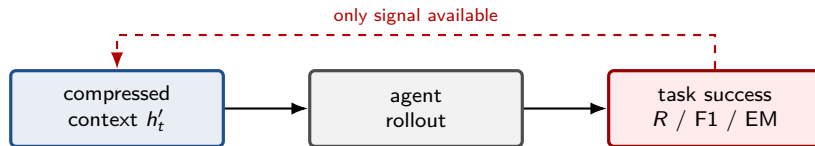
Why this number cannot be interpreted

I mixes two effects that change simultaneously:

- 1 genuinely composing f_o with f_h
- 2 the **shifted trigger schedule**

The two cannot be separated, so -9.5 is *not* evidence that composition itself is harmful.

Gap (c): Compression Quality Is Only Measured Indirectly



A failure here could mean either:
the compressor dropped a needed fact,
or the agent simply reasoned badly.

- ACON and SAC judge a compressor **solely** by downstream success.
- Neither measures whether the *specific facts the agent will need* survive compression.
- So a failure **cannot be attributed** to information loss rather than to the agent — the compressor is never diagnosed directly.

Problem Statement

Informally

Given a long-horizon agent with **frozen weights**, a history compressor and an observation compressor, determine how the two stages should be **composed and scheduled** so that the token savings of each are retained without lowering task success — holding the compression schedule fixed, and measuring compression quality *directly*.

Agent (POMDP, frozen θ):

$$a_t = M(h'_{t-1}, o'_t; \theta)$$

Compression operators:

$$o'_t = \begin{cases} f_o(o_t, h'_{t-1}) & |o_t| > T_{\text{obs}} \\ o_t & \text{else} \end{cases}$$
$$h'_t = \begin{cases} f_h(x_t) & t \in \mathcal{K} \\ h'_{t-1} \oplus (o'_t, a_t) & \text{else} \end{cases}$$

Two composition orders:

$$x_t^{\text{seq}} = h'_s \oplus (o'_{s+1}, a_{s+1}, \dots, o'_t, a_t)$$
$$x_t^{\text{dec}} = h'_s \oplus (o_{s+1}, a_{s+1}, \dots, o_t, a_t)$$

ACON uses x^{seq} .

Cost, peak context and latency:

$$C(\pi) = \sum_{t=1}^T (|h'_{t-1}| + |o'_t|)$$
$$L(\pi) = \sum_t \ell_M + \sum_{t \in \mathcal{K}} \ell_{f_h} + \sum_{|o_t| > T_{\text{obs}}} \ell_{f_o}$$

Formal objective:

$$\pi^* = \arg \max_{x, \mathcal{K}} \mathbb{E}[R(\pi)] - \lambda \mathbb{E}[C(\pi)] - \mu \mathbb{E}[L(\pi)]$$

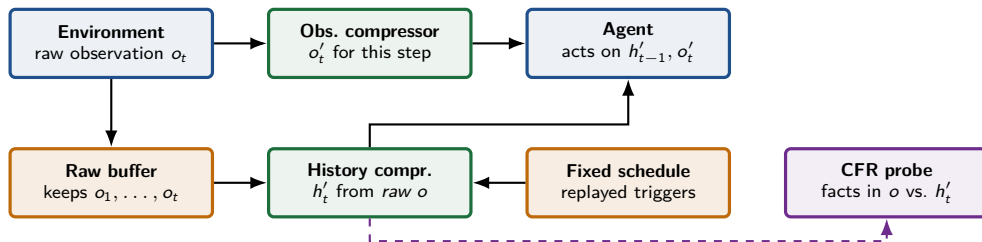
subject to

$$\mathbb{E}[R(\pi)] \geq \mathbb{E}[R(\pi_\emptyset)] - \varepsilon$$

$$C(\pi_{ho}) \leq \min\{C(\pi_h), C(\pi_o)\}$$

$$\mathcal{K}(\pi_{ho}) = \mathcal{K}(\pi_h) \text{ when comparing them}$$

Proposed Pipeline



1. Decoupled composition order

The agent still gets the short o'_t , but the **raw** o_t is kept in a side buffer and f_h receives x_t^{dec} . If f_o only deletes content then $V(o'_t) \subseteq V(o_t)$, so facts dropped by the observation stage *can still reach* the history summary.

2. Controlled compression schedule

Record $\mathcal{K}^h = \mathcal{K}(\pi_h)$ from the history-only run and **replay** it: $\mathcal{K}(\pi_{ho}) := \mathcal{K}^h$. With the schedule fixed, π_h and π_{ho} differ *only* in f_o , so I now measures composition alone.

3. Critical-Fact Retention: Measuring the Compressor Directly

Definition

Let $V(\cdot)$ extract state variables — access tokens, entity IDs, file paths, numeric values — and let $a_{1:T}^\emptyset$ be the actions of the **uncompressed reference run**. The facts at stake in trigger t , and the retention rate, are

$$F_t = V(x_t) \cap \bigcup_{t'} V(a_{t'}^\emptyset), \quad \text{CFR}(\pi) = \frac{\sum_{t \in \mathcal{K}} \sum_{v \in F_t} \mathbf{1}[v \sqsubseteq h'_t]}{\sum_{t \in \mathcal{K}} |F_t|},$$

where $v \sqsubseteq h'_t$ means an *exact, verbatim* occurrence of v in h'_t .

Why it is cheap

- $V(\cdot)$ is a set of pattern extractors for AppWorld's tokens, IDs, paths and numbers
- Computed **offline** for every compressor call
- **No additional agent rollouts** required

What it lets us test

- Rank correlation $\rho(\text{CFR}, R)$ — does fact retention actually predict task success?
- CFR under x^{seq} vs. x^{dec} — does decoupling really retain more?
- Attribute a failure to **information loss** rather than to the agent

Future Work

Complete the reproduction

- Finish the 38-task history-compression run
- Run observation and combined history + observation compression
- Add the paper thresholds (4096/1024) and the llama3.1:8b compressor on the same split

Implement the proposed pipeline

- Raw-observation buffer (x^{dec})
- Schedule replay $\mathcal{K}(\pi_{ho}) := \mathcal{K}^h$
- CFR extractor; then estimate I under *both* orders with the schedule held fixed

Measure latency directly

Log $\ell_M, \ell_{f_h}, \ell_{f_o}$ per call to decompose $L(\pi)$, including **prefix re-encoding** after each history compression.

Strengthen the evidence

- At least **three seeds** per setting
- Raise the served context window above 4096 tokens so baseline prompts are not truncated
- For SAC $\times$ ACON: run **TGA alone**, plus the 5 $\times$ and 51 $\times$ ratios

References

-  M. Kang, W.-N. Chen, D. Han, H. A. Inan, L. Wutschitz, Y. Chen, R. Sim, and S. Rajmohan, “ACON: Optimizing Context Compression for Long-horizon LLM Agents,” *ICML*, PMLR 306, 2026.
-  X. Liu, R. Zhao, P. Huang, X. Liu, J. Xiao, C. Xiao, T. Xiao, S. Gao, Z. Yu, and J. Zhu, “Autoencoding-Free Context Compression for LLMs via Contextual Semantic Anchors,” *ICLR*, 2026.
-  Y. Wang, Y. Shi, M. Yang, R. Zhang, S. He, H. Lian, Y. Chen, S. Ye, K. Cai, and X. Gu, “SWE-Pruner: Self-Adaptive Context Pruning for Coding Agents,” arXiv:2601.16746, 2026.
-  H. Jiang, Q. Wu, C.-Y. Lin, Y. Yang, and L. Qiu, “LLMLingua: Compressing Prompts for Accelerated Inference of Large Language Models,” *EMNLP*, pp. 13358–13376, 2023.
-  Z. Zhou et al., “MEM1: Learning to Synergize Memory and Reasoning for Efficient Long-Horizon Agents,” arXiv:2506.15841, 2025.
-  H. Yu et al., “MemAgent: Reshaping Long-Context LLM with Multi-Conv RL-based Memory Agent,” *ICLR*, 2026.
-  H. Trivedi et al., “AppWorld: A Controllable World of Apps and People for Benchmarking Interactive Coding Agents,” *ACL*, pp. 16022–16076, 2024.



Thank You

Questions?

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